INTRODUCTION TO BUSINESS ANALYTICS

POWER BI PRACTICAL – QUESTION BANK

SECTION 1 – Data Preparation using Power Query

1. Import the dataset into Power BI and open **Power Query Editor**.
2. Identify and remove duplicate records from the dataset.
3. Ensure correct **data types for all columns** (Date, Revenue, Quantity, Price, Region, etc.).
4. Create a new column: **Revenue = Price × Quantity**.
5. Extract **Month Name** from the Date column.
6. Extract **Year** from the Date column.
7. Extract **Day Name** from the Date column.
8. Create a **Sales Category column (Low / Medium / High)** based on Revenue using Conditional Column.
9. Rename columns properly and apply meaningful column names.
10. Extract **Quarter from Date** column.
11. Remove **blank or null rows** from the dataset.

SECTION 2 – Descriptive Analytics using DAX

Create the following DAX Measures:

1. Total Revenue
2. Total Quantity Sold
3. Average Revenue
4. Standard Deviation of Revenue
5. Maximum Revenue
6. Region-wise Total Revenue
7. Identify **Top 3 Sales_Person by Revenue using RANKX**
8. Create a measure to calculate **Minimum Revenue**.
9. Calculate **Total Number of Transactions (Count of records)**.

SECTION 3 – Dashboard Creation

Create a Power BI Dashboard including:

1. **Bar Chart – Sales by Region**
2. **Column Chart – Sales by Sales_Person**
3. **Line Chart – Sales Trend over Time**
4. Add at least **one slicer (Month or Region)**.
5. Add **KPI Cards for Total Revenue and Total Quantity Sold**.
6. Create a **Pie Chart showing revenue distribution by Region**.

SECTION 4 – Business Insights

Write insights based on your dashboard:

1. Identify the **best performing region**.
2. Identify the **best performing sales person**.
3. Describe the **monthly sales trend**.
4. Explain **revenue distribution across regions**.
5. Write an **overall performance summary of the business**.
6. Suggest **two strategies to improve sales performance**.
7. Identify **which region has potential for growth** based on the data.

POWER BI PRACTICAL – ANSWER KEY / SOLUTION GUIDE

📊 SECTION 1 – Data Preparation using Power Query

1. Import the dataset into Power BI

Steps

1. Open **Power BI Desktop**
2. Click **Get Data**
3. Select dataset type (Excel / CSV)
4. Click **Load**
5. Open **Transform Data** to enter Power Query Editor

2. Remove Duplicate Records

Steps

1. Open **Power Query Editor**
2. Select all columns
3. Click **Remove Rows**
4. Click **Remove Duplicates**

3. Ensure Correct Data Types

In Power Query, set the following types:

Column	Data Type
Date	Date
Price	Decimal Number
Quantity	Whole Number
Revenue	Decimal Number
Region	Text
Sales_Person	Text

Steps

Select column → Change Data Type.

4. Create Revenue Column

If revenue not given:

Power Query → **Add Column → Custom Column**

Formula:

```
Revenue = [Price] * [Quantity]
```

5. Extract Month Name

Power Query → **Add Column → Date → Month → Month Name**

OR DAX:

```
Month = FORMAT(Sales[Date], "MMMM")
```

6. Extract Year

Power Query:

Add Column → Date → Year

OR DAX:

```
Year = YEAR(Sales[Date])
```

7. Extract Day Name

Power Query:

Add Column → Date → Day → Name of Day
OR
```
Day_Name = FORMAT(Sales[Date], "dddd")
```

8. Create Sales Category Column

Power Query → **Add Column → Conditional Column**
Example rule:

Condition	Output
Revenue < 1000	Low
1000–5000	Medium
5000 | High |	

9. Rename Columns

Right-click column → **Rename**
Example:

Old Name	New Name
qty	Quantity
rev	Revenue
sp	Sales_Person

10. Extract Quarter

Power Query:
Add Column → Date → Quarter → Quarter of Year
OR
```
Quarter = "Q" & FORMAT(Sales[Date], "Q")
```

11. Remove Blank Rows

Power Query → **Remove Rows → Remove Blank Rows**

📈 SECTION 2 – Descriptive Analytics using DAX

1. Total Revenue

```
Total Revenue = SUM(Sales[Revenue])
```

2. Total Quantity Sold

```
Total Quantity = SUM(Sales[Quantity])
```

3. Average Revenue

```
Average Revenue = AVERAGE(Sales[Revenue])
```

4. Standard Deviation

```
Std Dev Revenue = STDEV.P(Sales[Revenue])
```

5. Maximum Revenue

```
Max Revenue = MAX(Sales[Revenue])
```

6. Region-wise Revenue

Use **Bar Chart**
Axis → Region
Values → Total Revenue

7. Top 3 Sales_Person by Revenue

Create Rank Measure:

```
Sales Rank =
RANKX(
ALL(Sales[Sales_Person]),
[Total Revenue],
,
DESC
)
```

Then apply **Top N Filter = 3**

8. Minimum Revenue

```
Min Revenue = MIN(Sales[Revenue])
```

9. Total Transactions

```
Total Transactions = COUNTROWS(Sales)
```

📊 SECTION 3 – Dashboard Creation

1. Bar Chart – Sales by Region

Visual: **Bar Chart**

Axis → Region

Values → Total Revenue

2. Column Chart – Sales by Sales_Person

Axis → Sales_Person

Values → Total Revenue

3. Line Chart – Sales Trend

Axis → Date

Values → Total Revenue

4. Add Slicer

Insert **Slicer**

Field options:

- Month
- Region

5. KPI Cards

Insert **Card Visual**

Measures:

```
Total Revenue
Total Quantity
Average Revenue
```

6. Pie Chart – Revenue Distribution

Visual: **Pie Chart**

Legend → Region

Values → Total Revenue

📌 SECTION 4 – Business Insights

Example answers students may write:

1. Best Performing Region

Region with **highest total revenue**.

Example:

North region contributes the highest revenue compared to other regions.

2. Best Performing Sales Person

Sales person with **maximum total sales**.

Example:

John generated the highest sales revenue.

3. Monthly Sales Trend

Based on line chart.

Example:

Sales show increasing trend in the middle months of the year.

4. Revenue Distribution

Example:

Majority of revenue comes from two regions, indicating market concentration.

5. Overall Performance

Example:

The business shows stable growth with moderate variation across regions.

6. Strategies to Improve Sales

Example:

- Expand marketing in low-performing regions.
- Provide incentives to sales team.

7. Growth Opportunity

Example:

Regions with increasing monthly sales trend have potential for expansion.